

Peeling the first coordinate of a Bochner box integral

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Abstract

Infrastructure for the next geometric bridges (Astra #17: tangential integration, exponential localisation, symmetric normal boxes). The $[0, \infty]$ -valued peel of unit 174 gets a Bochner counterpart for an arbitrary measurable one-dimensional factor S and an integrable integrand: $\int_{S^{d+1}} F = \int_S \int_{S^d} F(\text{cons}(a, b)) db da$ (`integral_pi_box_succ`). Also in this unit: the independent statement-level review v4 of units 184–187 (no blocking findings; two should-fix items applied) and Astra #17 saved. Zero `sorry/axiom`.

1 The things formalised

```
def piBox (d : ℕ) (S : Set α) : Set (Fin d → α) := Set.pi univ fun _ => S
theorem restrict_piBox (d S) :
  volume.restrict (piBox d S) = Measure.pi fun _ => volume.restrict S
theorem integral_piBox_zero (S) (F) : ∫ x in piBox 0 S, F x = F.isEmptyElim
theorem integral_pi_box_succ (d) (S) (F) (hF : IntegrableOn F (piBox (d + 1) S)) :
  ∫ x in piBox (d + 1) S, F x = ∫ a in S, ∫ b in piBox d S, F (Fin.cons a b)
```

2 Review v4 and Astra #17

Review v4 (`tide-log/gpt6_fidelity_review_v4.md`) recomputed the sanity example ($k = (1, 1)$, $h = (0, 2)$, $\eta = 1 + y$: $N^{1/2}I_\eta \rightarrow 5\sqrt{\pi}/12$) from the Lean statement, confirmed the mirror equation, and asked for two changes, both applied: `amplitudeCoeff_eq_zero_of_face` now only requires vanishing on the face *over the box*, and the equal-ratio prose is qualified (origin evaluation holds for *every* continuous amplitude exactly when all ratios are equal; particular amplitudes satisfy it otherwise). Astra #17 (`gpt6_bigpicture_v17.md`) ranks tangential integration first (average the amplitude over the tangential variable, then apply the amplitude theorem once), then a local leading-density headline, exponential-gap localisation, and symmetric normal boxes with signed reflections; and recommends an author-facing consolidated report, which accompanies this unit in the staging area.

3 Lean notes

- `MeasurePreserving.integral_comp` and `integrable_comp_emb` transport both the integral and the integrability along `piFinSuccAbove`; the Bochner Fubini `integral_prod` needs the transported integrability.
- The `IntegrableOn` hypothesis is unfolded and rewritten with `restrict_piBox` before the transport, so that the product measure appears syntactically.